STUDENT REFERENCE | BACKGROUND FIRST, HISTORICAL REVEAL SECOND

What is measles?

Disease, prevention, vaccine history, and the Edmonston–Zagreb (EZ) research controversy

AT A GLANCE

Measles (rubeola) is a highly contagious viral infection transmitted through the air. It is not the same illness as rubella (German measles). A person with measles can infect others before the characteristic rash appears.

The infection

- Cause and spread: Measles virus (a morbillivirus) spreads through airborne particles and respiratory secretions. Infectious virus can remain in indoor air for up to two hours after an infected person leaves.
- Contagious period: Typically from four days before to four days after rash onset. Up to 9 of 10 susceptible close contacts may become infected.
- Symptoms: High fever; cough, runny nose, and red eyes; sometimes Koplik spots inside the mouth; followed by a rash that usually starts on the face and moves downward.
- Complications: Ear infection and diarrhea are common; pneumonia, encephalitis, and death can occur. Infants, young children, pregnant people, and people with weakened immune systems face particular risks.

Why vaccination matters

Before measles vaccines, an estimated 3–4 million U.S. infections occurred each year. In the decade before vaccine introduction, an estimated 400–500 people died each year, and tens of thousands were hospitalized. [1, 2]

Current U.S. vaccines contain live, attenuated (weakened) measles virus. CDC estimates about 93% protection against measles after one MMR dose and 97% after two doses. Routine childhood protection uses a two-dose schedule; circumstances such as international travel and outbreaks may call for earlier vaccination. [3]

ONE IMPORTANT DISTINCTION

A vaccine can be safe and effective in routine use while an unlicensed formulation, different dose, younger age, or experimental schedule still requires its own evidence and appropriate research consent.

Discussion starter

Suppose an outbreak is disproportionately affecting infants too young for routine vaccination. What would make an early-vaccination research study scientifically useful? What would make it ethically acceptable?

Sources for this page: [1] CDC, History of Measles; [2] CDC, Clinical Overview of Measles; [3] CDC, Measles Vaccine Recommendations. Full links on page 2.

HISTORICAL CONTEXT | REVEAL WITH STAGE 5; CONTAINS SPOILERS

Where EZ fits in vaccine history

U.S.-licensed products versus an investigational vaccine strategy

Year	U.S. measles vaccine milestone
1963	First U.S. licenses: live attenuated Edmonston B and inactivated (killed) measles vaccines.
1965–1968	Further-attenuated Schwarz strain introduced (1965); inactivated vaccine withdrawn (1967); Edmonston–Enders / Moraten licensed (1968).
1971	Combined measles–mumps–rubella (MMR) vaccine licensed.
1989	Recommendations for a routine second measles-containing vaccine dose emerged after outbreaks.
2005	Combined measles–mumps–rubella–varicella vaccine (MMRV; ProQuad) licensed.
2022	A second MMR product, PRIORIX, licensed in the United States; M-M-R II and PRIORIX are interchangeable for recommended MMR use.

What was Edmonston–Zagreb (EZ)?

EZ is a live-attenuated measles vaccine strain used outside the United States. It was not a U.S.-licensed measles vaccine product. Researchers studied whether higher-potency EZ formulations could protect infants earlier than the routine U.S. schedule. “High-titer” describes the viral dose/potency, not a completely different measles virus disease. It is essential to distinguish high-titer investigational regimens from standard-titer EZ vaccination outside the United States and from routinely licensed U.S. MMR products. [4–6]

The Los Angeles study: what the record supports

During the 1989–1991 U.S. measles epidemic, a CDC-supported study enrolled approximately 1,500 Los Angeles infants beginning in 1990. Children were assigned among five vaccine study groups involving EZ formulations or the U.S.-licensed Moraten vaccine. The research was reviewed by institutional review boards. Parents signed research consent documents, but the forms did not clearly disclose that EZ was experimental/unlicensed in the United States and inadequately described foreseeable risks and alternatives. [4]

Investigators stopped EZ administration in Los Angeles in October 1991 after concerning findings from Senegal and Haiti about increased mortality among girls given high-potency measles vaccine. WHO withdrew its recommendation for high-titer measles vaccine in 1992. Follow-up of the Los Angeles participants found no significant health problems attributed to the high-potency EZ vaccine. A federal review judged the study scientifically and ethically justified in its aims but identified deficient consent. The institutions notified families and apologized in 1996. [4–6]

DO NOT CONFLATE THESE FACTS

Safety signals from other countries were real and demanded action. They are not evidence that the Los Angeles trial caused vaccine-attributable deaths, nor do they establish that standard U.S. MMR vaccination carries the same risk.

Source notes and further reading

[1] CDC, History of Measles. [Open source]
[2] CDC, Clinical Overview of Measles. [Open source]
[3] CDC, Measles Vaccine Recommendations. [Open source]
[4] David Satcher testimony, U.S. House hearing, May 8, 1997, pp. 21–23. [Open source]
[5] WHO, 1992, Safety of High Titre Measles Vaccines. [Open source]
[6] CDC Pink Book, Ch. 13; PRIORIX MMWR 2022. [Open source]
[7] CDC, PRIORIX licensure and recommendations. [Open source]